

2023

PHYSICS — HONOURS — PRACTICAL

Paper : SEC-1P

Full Marks : 100

*The figures in the margin indicate full marks.***SET - 1****[LNB - 20, VIVA - 20, PROGRAMMING - 60]**Answer *any one* question from *each group*.**Group - A**

1. Three functions are defined in the given range as the following :

$$f(x) = 1.45x^3 - 6\sqrt{x}, \quad x \geq 2$$

$$g(x) = \sin^2(x), \quad -1 \leq x < 2$$

$$h(x) = |x|e^x, \quad x < -1$$

Set the range of x values from -2 to $+3$ and the range of y values from -5 to $+5$. Set ticks of the x -axis at an interval of 1.0 and ticks of y -axis at an interval 2.0 . Set title 'Piece-wise continuous functions'. Plot the three functions using GNUPLOT and store the plot in suitable format. 20

2. Plot the following functions for $0 \leq x \leq \pi$ on the same graph using GNUPLOT. Set 1000 samples. Mark each curve with different line style and line widths. Label x -axis and y -axis as 'x-values' and 'y-values' respectively. Set the title of the plot 'Function plots'. Save the plot in a suitable format.

$$f_1(x) = \sin^2(\sqrt{x}), \quad f_2(x) = e^{-x/2} \cos(2\pi x), \quad f_3(x) = 12.34x^{5/3}$$

20

3. Plot the following functions for $-10 \leq x \leq 10$ on the same graph using GNUPLOT. Set 1000 samples. Take the range of y -values from -4 to $+4$. Mark each curve with different line style and line widths. Label x -axis and y -axis as 'x-values' and 'y-values' respectively. Set the title of the plot 'Function plots'. Save the plot in a suitable format.

$$f_1(x) = \coth(x) - \frac{1}{x}, \quad f_2(x) = e^{-\sin(x)}, \quad f_3(x) = x|x|$$

20

4. Plot the following parametric functions using GNUPLOT :

$$x(r, t) = r(t) \cos(t), \quad y(r, t) = r(t) \sin(t),$$

where $r(t) = 3(1 + \cos(t))$. Set the range of t values from $-\pi$ to π . Set the title of the plot as 'Cardioid function'. Set samples 1000. Save the plot in a suitable format. 20

5. Plot the following polar equations in GNUPLOT :

$$r(\theta) = \theta^2 \sin(\theta/\pi), \quad r(\theta) = 3.75e^{-\frac{\theta}{2.39}}$$

Consider θ range to be $-\pi$ to π and the radial (r) range to be $-3 < r < 3$. Set different line types for the two functions. Put the title "Polar function plot". Save the plot in a suitable format. 20

6. Save the following data table in a file. Fit the data points of the second row against that of

x	-3.0	-2.0	-1.0	0.0	1.0	2.0	3.0	4.0
$f(x)$	3.422	4.057	4.808	5.7	6.756	8.008	9.492	11.251

the first row using GNUPLOT with the equation $f(x) = ae^{bx}$. Write the values of the fitted parameters a, b . Show the fitted curve along with the data points in the same plot. Set the title "Data fitting" and 'x' and 'f(x)' for x-label and y-label respectively. Save the plot in a suitable format. 20

7. Create the following data table in a file :

x	0.0	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
y	0.0	0.006	0.053	0.195	0.49	1.0	1.792	2.935	4.5	6.56	9.19

Fit the above data table with $f(x) = x^a$ with parameter a . Write the value of a . Plot the above data table along with $f(x)$.

Set the title 'Data fitting' and 'x' and 'f(x)' for x-label and y-label respectively. Save the plot in a suitable format. 20

Group - B

8. (a) Write a python program to find the root of the transcendental equation correct up to three decimal places $\sin(x) = \ln(x)$ in between $[0.001, \pi]$ using Bisection method.
 (b) Write a python program to calculate the sum of first 20 natural numbers. Print the output.

30+10

9. (a) Write a python program to find the velocity of an object at $t = 10$ s by solving the following differential equation using Euler method :

$$\frac{dv}{dt} = 3t^2 - \sqrt{t} - \cos(6t).$$

The initial velocity of the object is given by $v(t = 0) = 5 \text{ m / s}$.

Create a list of t -values with 1000 elements between $[0, 10]$. Put the y -values in a list Y . Print the last element of Y .

- (b) Consider the following matrix :

$$A = \begin{pmatrix} 1 & 4 & -2 \\ 2 & -1 & 3 \\ -4 & -3 & 1 \end{pmatrix}$$

Write a python program to find the transpose of matrix A . Print the new matrix.

30+10

10. (a) Using Newton-Raphson method with starting point $x_0 = 1$, determine the root of the following equation correct up to three decimal places :

$$x^3 - 2x^2 - 5x + 8 = 0.$$

Print the root.

- (b) Write a python program to calculate the sum of the following G.P. series :

$S = 2 + 6 + 18 + \dots$ up to 5 terms.

Print the output.

30+10

11. (a) Using Newton-Raphson method with starting point $x_0 = 1$, determine the root of the following equation correct up to three decimal places :

$$x \cos x - x^2 = 0.$$

Print the root.

- (b) Consider the following two matrices :

$$A = \begin{pmatrix} 3 & 4 \\ 5 & 6 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 0 \\ 1 & -2 \end{pmatrix}$$

Write a python program to find $A + B$. Print the resultant matrix.

30+10

12. (a) A first order differential equation is given as below :

$$\frac{dy}{dx} = 4x - 2 \quad \text{with } y(0) = 0.$$

Create a list (say X) with 100 x -values between $[0, 1]$. Solve this differential equation using Euler method for x -values in the list X . It is given that the exact solution is $y(x) = 2x^2 - 2$. Calculate the y -value for the last element of the list X . Print this value and compare with the value from the exact solution.

- (b) Create a list with following numbers :

27, 11, 5, 18.

Find the even and odd numbers and print accordingly.

30+10

13. (a) Write a python program to compute the exponential series up to three decimal places of accuracy :

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

Take $x = 1$ as input. Print the value of $\ln(e)$ using the computed value of e .

- (b) Write a python program to calculate and print the trace of the following matrix :

$$\begin{pmatrix} -7 & 2 & 1 \\ -3 & 4 & 8 \\ 5 & -1 & 2 \end{pmatrix}$$

30+10

14. (a) Calculate the sum of the following convergent series for $n = 2$ correct up to 5 decimal places :

$$1 + \frac{1}{n} + \frac{1}{n^2} + \frac{1}{n^3} + \dots$$

Verify that the sum is close to $\frac{n}{n-1}$.

- (b) The sequence F_n of Fibonacci numbers is defined by the recurrence relation :

$F_n = F_{n-1} + F_{n-2}$ with seed values and $F_0 = 0$ and $F_1 = 1$. Calculate and print the 10th term of this series.

30+10